This has created a scientific area called *elementary particle physics*. It is quite different from nuclear physics, which is concerned with composite nuclei only. A principal objective of elementary particle physics has been to group the particles together according to their properties to obtain a meaningful pattern, which would describe all particles as parts of some few fundamental building blocks of nature. One step in this direction is to study how the elementary particles interact with each other, i.e. what kind of forces are involved.

Scientists have long doubted that all the particles produced with masses between the electron and the proton (loosely referred to as *mesons*, i.e. "intermediate"), and with masses greater than the proton (referred to as *baryons*, "heavy") really are "elementary". It was proposed that they have a substructure or constitute excited states of each other. Are they waves or particles since they serve as carriers of force? At this point it is important to understand what is meant by "particle" in nuclear physics.

2.2. FORCES OF NATURE

Considering what an immense and incredible diverse assembly the universe is — from the cosmos to man and microbes — it is remarkable that scientists have been able to discover with certainty only four basic forces, which govern the attraction and repulsion of all physical objects of nature. Let us consider these *forces of nature* in a qualitative way from the weakest to the strongest, see Table 2.1.

The first and weakest force of nature is that of *gravity*. The gravitational force, F_g, beween two masses, m_1 and m_2, at a distance r is given by eqn (2.1), where **G** is the universal constant of gravity ($\mathbf{G} = (6{,}6742 \pm 0{,}0007) \times 10^{-11}$ N m^2 kg^{-2}).

$$F_g = \mathbf{G} \times m_1 \times m_2 / r^2 \tag{2.1}$$

F_g is the force that causes all objects to always attract one another and is responsible for the attraction of the planets to the sun in the solar system and of the solar system to the rest of the galaxy. It is also the force that holds us to the earth. It seems paradoxical that

Table 2.1 Forces of nature in order of strength and their exchange particles

Force	Relative strength	Range	Exchange particle	Spin	Rest mass
1. Gravitation	1	very long	graviton[†]	2	0
2. The weak force	10^{25}	very short	W^+, W^-	1	79.9 GeV
			Z^0	1	91.2 GeV
3. The electromagnetic force	10^{36}	long	photon	1	0
4. The nuclear force (the strong force)	10^{38}	10^{-14} m	gluon	1	0

[†]Not yet detected.

the weakest attraction of the four basic forces of nature is the force that is responsible for the assembly of the largest objects on the greatest scale. In modern physics it is believed that all forces are carried by "something" which either can be described as a wave or as a particle (see §2.3). The carrier of the gravitational force is the *graviton*. Experimenters have tried various ways to detect gravitational waves, but so far the results are negative.

The second weakest force is the one which is involved in the radioactive β-decay of atoms (see Ch. 5). It is known as the *weak interaction force*. This weak interaction force operates over extremely short distances and is the force that is involved in the interaction of very light particles known as *leptons* (electrons, muons, and neutrinos) with each other and as well as their interaction with mesons, baryons, and nuclei. The carrier of the weak interaction force is still a matter of considerable research; we will return to this point later.

The third weakest force of nature with which we are all relatively familiar is that of the *electromagnetic force*. The electromagnetic force, F_e, is expressed by Coulomb's law and is responsible for the attraction and repulsion between charged bodies, see eqn (2.2) where z_1 and z_2 are the charges, r is the distance and $\mathbf{k}_e$ is a constant ($\mathbf{k}_e = 1/(4\pi\varepsilon_0)$ in the SI-system).

$$F_e = -\mathbf{k}_e \times z_1 \times z_2 / r^2 \tag{2.2}$$

Just as the gravitational force holds the planets in their orbits about the sun and explains the stability of the planetary systems, the electromagnetic force explains the attraction/repulsion between electrons in atoms, ionic bonds in molecules, molecules and ions in crystals. It is the force that holds the atomic world together. It is approximately 10^{36} times stronger than the gravitational force. If gravity is the force underlying the laws of astronomy, electromagnetism is the force underlying the laws of chemistry and biology. The carrier of the electromagnetic force is the *photon*.

The fourth major force in nature is the nuclear force. This force is also known as the *strong interaction force* and is the one responsible for holding nuclear particles together. Undoubtedly it is the strongest in nature but operates only over the very short distance of approximately 10^{-14} m and saturates rapidly, i.e. only a few nucleons are involved. Whereas electromagnetism binds electrons to nuclei in atoms with an energy corresponding to a few electron volts, the strong interaction force holds nucleons together in nuclei with energies corresponding to millions of electron volts. The carrier of the strong interaction force is recognized to be the *gluon*; we will return to this point in §2.6. One characteristic of leptons is that they seem to be quite immune to the strong interaction force.

The general comparison between the different forces is shown in Table 2.1. The strong nuclear force is approximately 10^2 times greater than the Coulombic force, while the weak interaction force is smaller than the strong attraction by a factor of approximately 10^{13}.

There is also a considerable difference in the reaction time of the different interactions. The strong interaction manifests itself in its ability to react in very short times. For example, for a particle, which passes an atomic nucleus of about 10^{-15} m in diameter with a velocity of approximately 10^{8} m s^{-1} (i.e. with a kinetic energy of ~50 MeV for a proton and 0.03 MeV for an electron), the time of strong interaction is about 10^{-23} s. This is about the time of rotation of the atomic nucleus. The weak interaction force requires a much longer reaction time and explains why leptons such as electrons and photons do not react with atomic nuclei but do react with the electron cloud of the atom, which has a diameter on the order of 10^{-10} m. There is sufficient time in passing this larger diameter for the weak interaction force to be effective. It is expected that the electromagnetic force and the gravitational force are very fast compared to the weak nuclear force.

2.3. WAVES AND PARTICLES

It is daily experience that moving bodies have a kinetic energy, E_v, which involves a mass and a velocity, v. Less familiar is the concept of Planck (1900) in which light moves in wave packets of energy:

$$E = \mathbf{h}\nu \tag{2.3}$$

Here ν is the frequency of light with wavelength λ

$$\mathbf{c} = \nu\lambda \tag{2.4}$$

and **h** the Planck constant, 6.63×10^{-34} J s.

Einstein — in the theory of the photoelectric effect —, and Compton — in the theory of the scattering of photons (Ch. 6) — showed that photons have not only a discrete energy, but also a discrete momentum (p_v).

$$p_v = E_v/\mathbf{c} \tag{2.5}$$

Photons seem to collide with other particles as if they have a real mass and velocity as in the classical mechanical expression for momentum: $p = mv$. If we put $v = \mathbf{c}$ and equate with (2.5) we obtain a relativistic mass of the photon as

$$m_v = E_v/\mathbf{c}^2 \tag{2.6}$$

This is the mass-energy relation of Einstein.

There are many examples of mass properties of photons. To the two mentioned above we may add the solar pressure (i.e. photons from the sun which push atoms away from the sun and into space), which has played a significant part in the formation of our planetary system, and measurements showing that photons are attracted by large masses through the gravitational force. Thus we see the evidence for the statement in the beginning that all moving elementary particles must have relativistic mass, even if the rest mass is zero.